

Fourier-Transform Decoding of Periodic Neural Dynamics in Zebrafish Calcium Imaging

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Abstract

A critical first step in analyzing calcium imaging datasets is identifying stimulus- or motor-related neurons, typically achieved through regression-based methods. However, these approaches often fail to detect transiently active phasic neurons. In this study, we present a complementary analysis pipeline based on Fourier transformation and apply it to a public calcium imaging dataset of zebrafish larvae undergoing phototactic stimulation. Our method reveals spatiotemporal patterns of neuronal activity across multiple brain regions while successfully capturing transient responses that regression analysis misses. Validation against conventional regression results confirms the robustness of our approach. Although the requirement for periodic perturbation in neural activity, Fourier analysis proves to be a powerful addition to the computational neuroscience toolkit, offering unique insights on temporal patterns of neural dynamics when used alongside existing analytical techniques.

Introduction

In recent years, the construction of large-scale neuronal activity datasets has become increasingly common in neuroscience. Advances in functional imaging technologies have enabled whole-brain recordings of neuronal activity dynamics at single-cell resolution¹. However, the richness of these datasets presents a challenge for neuroscientists, who must develop analytical tools to extract meaningful insights from neuronal activity data. A prominent example is the field of calcium imaging utilizing zebrafish. Over the past decade, zebrafish (*Danio rerio*) have gained popularity as a model organism in computational neuroscience due to their body transparency, well-established genetic toolkits, and extensively characterized behavioral responses. These features facilitate the study of neuronal activity in behaving animals^{2,3}.

Typically, the analysis of such neuronal datasets begins with identifying neurons associated with specific stimuli or motor responses, often achieved through regression analysis. However, regression-based methods have limitations: they struggle to label neurons whose activity patterns deviate from stimulus inputs or motor outputs and are insensitive to variations in neuronal activity strength. In this study, we propose an alternative analytical pipeline based on Fourier transformation, which can be applied to neuronal data under periodic perturbations.

We evaluated this Fourier-based analysis pipeline using a publicly available zebrafish calcium imaging dataset⁴, which comprehensively examined innate sensorimotor behaviors—including phototaxis—through regression analysis. Phototaxis, the tendency of zebrafish to orient toward illumination⁵, was elicited in this dataset by a periodic half-field dark stimulus.

The periodicity of the stimulus makes it suitable for Fourier analysis.

In this study, we first outline the mechanism and schematic of our analytical pipeline based on the Discrete Fourier Transform (DFT). Next, we present an anatomical map derived from Fourier analysis and compare it with existing findings in the original research using this datasets⁴. Finally, we contrast our Fourier analysis results with the regression-based results from the original dataset to assess the validity of our approach. By elucidating the principles of Fourier analysis and comparing its advantages and limitations to established methods, we aim to demonstrate its utility as an alternative framework for extracting insights from neuronal activity datasets.

Result

Fourier analysis identifies stimulus-locked neurons by extracting periodic components

In this study, we analyzed whole-brain neuronal activity data from 10 zebrafish larvae (*Danio rerio*) subjected to whole-field visual stimulation using light-sheet microscopy. The dataset was originally published and made publicly available as a study of neuronal mechanism underlying sensorimotor behavior⁴. Our analysis focused specifically on phototaxis - an innate sensorimotor response characterized by orientation toward brighter regions of the visual field⁵. This behavioral response was reliably evoked using a stimulus paradigm consisting of simultaneous bright half-field and dark half-field illumination presented on opposite sides.

As described in the source of the dataset⁴, the phototaxis paradigm employed alternating left/right presentations of dark half-field stimulus against a bright background, effectively directing fish orientation toward the illuminated side (e.g., a left phototactic stimulus combined right dark half-field block with left bright half-field block). These stimulus blocks were interspersed with whole-field bright baseline periods. Neural activity was recorded at approximately 2 Hz frame rate during periodic phototaxis stimulation. Each imaging session captured activity from approximately 100,000 neurons per fish, with recording durations ranging from ~5 minutes to ~30 minutes across samples.

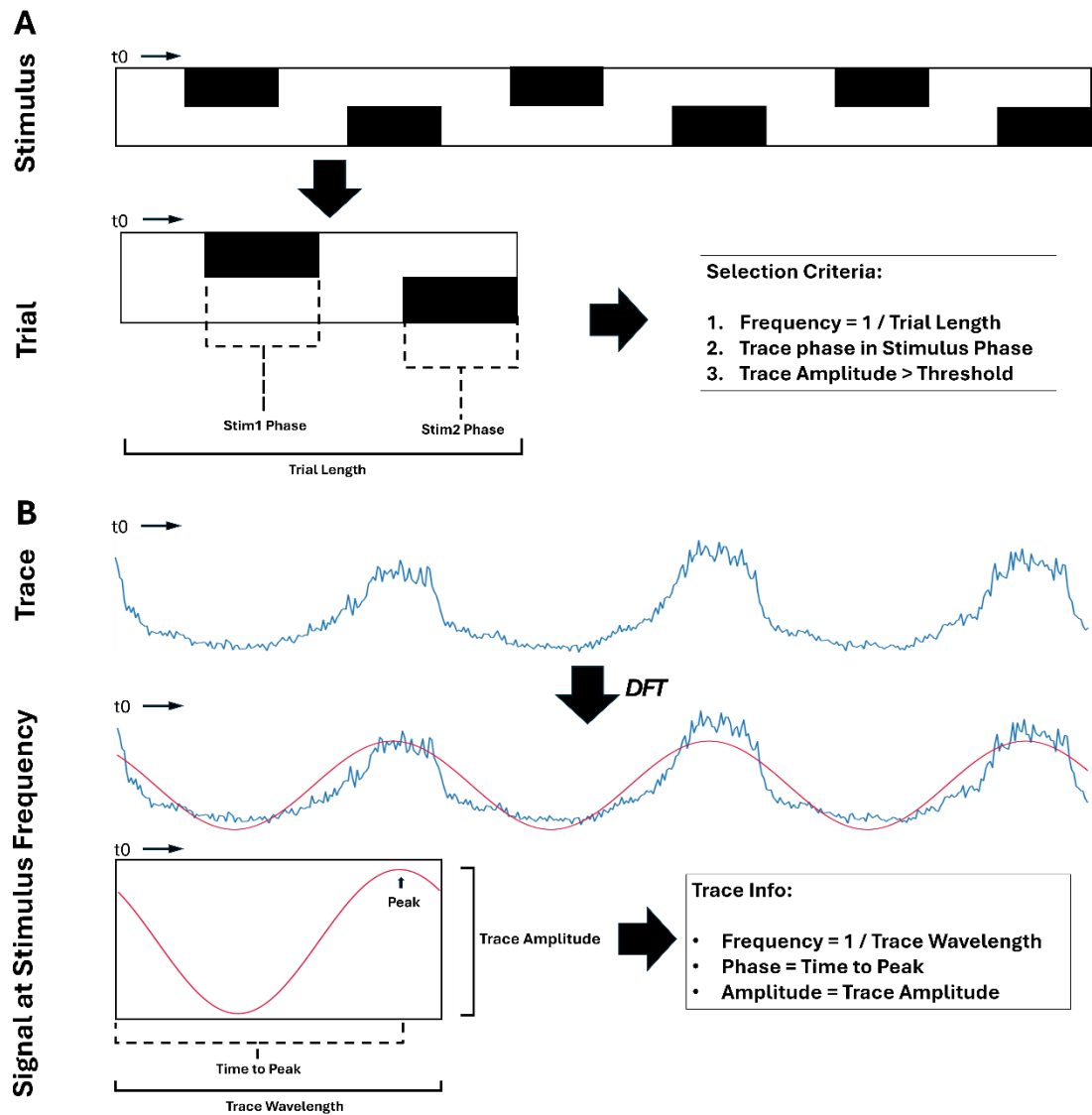


Figure 1. | Schematic of Identifying Stimulus-Related Neuron based on Discrete Fourier Transformation (DFT). (A) Schematic of periodic phototactic stimulus (left/right alternation) and derived selection criteria. **(B)** Discrete Fourier Transformation (DFT) of example neuronal trace showing component at stimulus frequency.

The periodic nature of the phototactic stimulus allowed segmentation into discrete, identical stimulation trials of equal duration and pattern (Figure 1A). In our experimental paradigm, each trial comprised two stimulus blocks of equal duration - one presented to each lateral visual field - separated by a bright baseline interval. This design enabled the stimulus to be represented as a periodic signal with frequency $F = 1/\text{trial duration}$. We further decomposed the bilateral stimulus into two unilateral components, each encoding phototactic stimulation from a single hemifield. Consequently, the complete stimulus can be represented as the superposition of two periodic signals sharing the same frequency but differing in phase (temporal offset of stimulus onset). This phase difference proves particularly useful when examining the laterality of stimulus-driven neuronal responses.

We used Discrete Fourier Transformation (DFT) to decompose neuronal activity during periodic phototactic stimulation into periodic components characterized by frequency, amplitude, and phase. These three parameters were systematically compared with the stimulus properties to identify sensory-responsive neurons. For neurons exhibiting stimulus-driven activity, we expected to observe a dominant periodic component matching the known stimulus frequency in their activity patterns. Stimulus-driven neurons were identified by isolating the periodic component corresponding precisely to the stimulus frequency (Figure 1B). The DFT provides a sinusoidal approximation of activity at this specific frequency, where the frequency parameter is predetermined by our analysis pipeline, the amplitude reflects the response strength as measured by fluorescence intensity changes, and the phase indicates the temporal alignment of peak response relative to beginning of stimulus trial. The amplitude parameter serves as a quantitative measure of neuronal response magnitude, with larger amplitudes corresponding to more pronounced fluorescence transients and thus stronger activation. We validated this relationship by demonstrating that neuron activity with higher amplitude components showed greater correlation to stimulus regressor in reported regression analysis⁴ (Figure 8C). The phase parameter, originally expressed in radians, was converted to imaging frames for more intuitive interpretation (where phase value n indicates peak response occurring at the n -th frame). This conversion facilitates direct comparison between the temporal dynamics of neuronal responses and stimulus presentation.

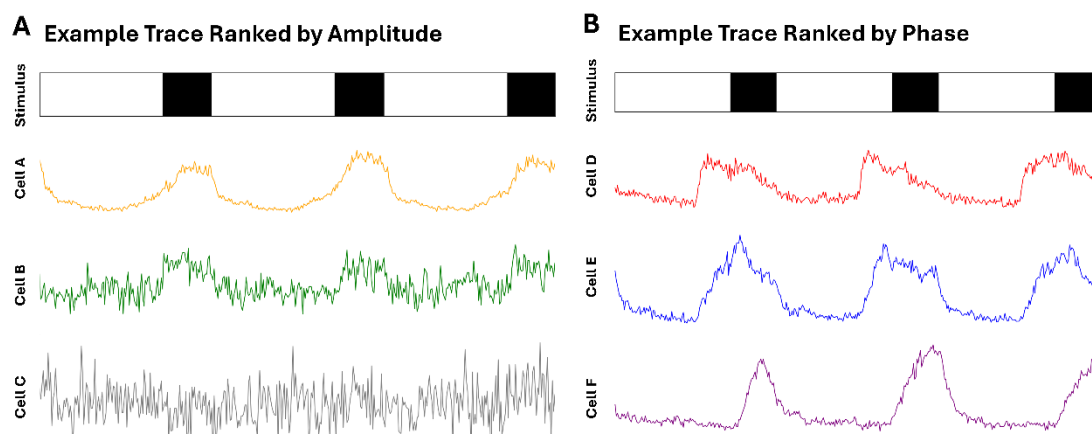


Figure 2. | Example of Neurons with Different Amplitude and Phase. (A) Example neurons (Cell A-C) sorted by descending Fourier amplitude ($\Delta F/F$). Neural activity is aligned left-side phototactic stimulation (3 trials). **(B)** Example neurons (Cell D-F) sorted by ascending phase (stimulus-relative timing) during left-side phototactic stimulation. Neural activity is aligned with left-side phototactic stimulation (3 trials)

Together, this approach yields two fundamental metrics for characterizing phototaxis-related neurons: the amplitude quantifies response strength (Figure 2A), while the phase encodes precise response timing relative to the stimulus (Figure 2B). The combination of these measures provides a comprehensive description of how individual neurons participate in the phototaxis circuit, capturing both the intensity and temporal coordination of their responses to periodic visual stimulation.

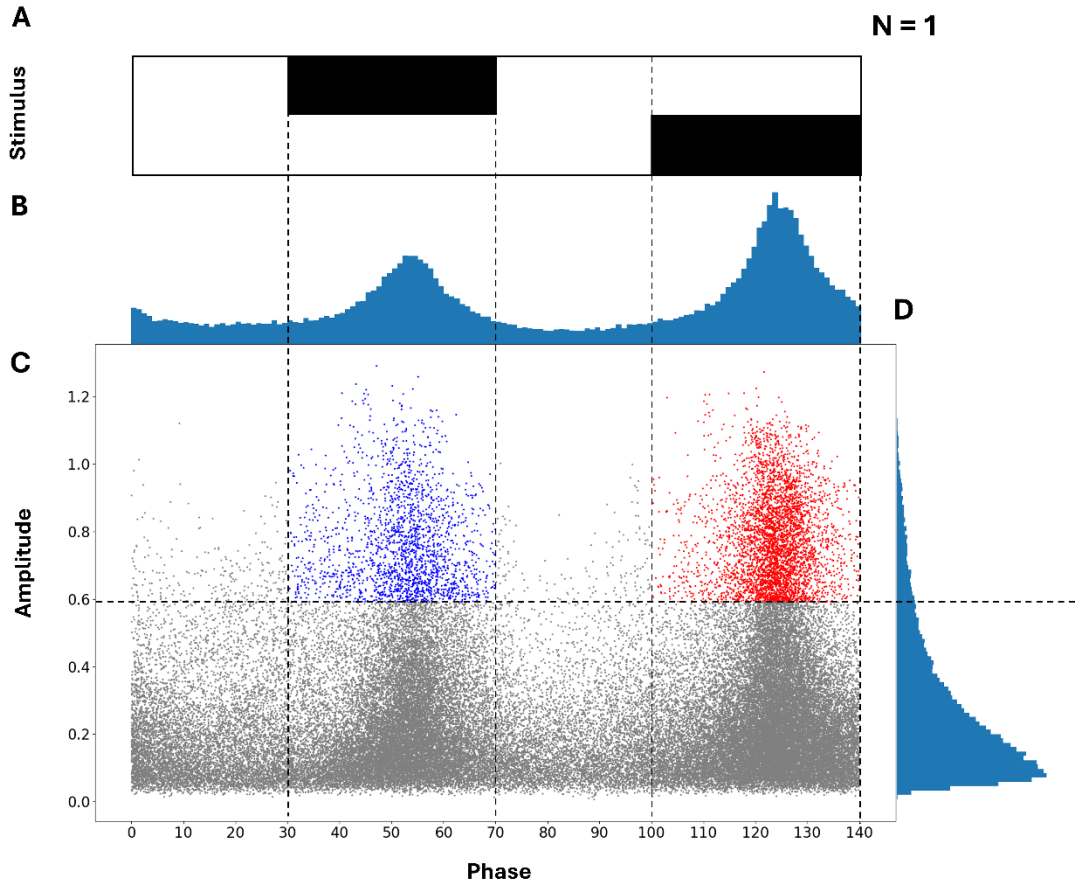


Figure 3. | Identification of Sensory-Related Neuron based on Amplitude-Phase space in one Example Fish. (A) Single trial showing right-side (frames 30-70) and left-side (frames 100-140) phototactic stimulation aligned to neuronal phase values. (B) Phase distribution histogram of neurons containing a stimulus-locked components ($n = 66,789$). (C) Amplitude-phase scatter plot showing neurons associated with right-side (blue) and left-side (red) phototactic stimulation, with selection cutoffs (horizontal dashed lines: top 10% amplitude cutoff; vertical dashed line: phase cutoff). (D) Amplitude distribution histogram of all detected neurons.

We implemented a two-stage selection process to identify neurons exhibiting both strong and temporally precise responses to phototactic stimulation. First, neurons were ranked by the amplitude of their stimulus-frequency component, with the top 10% selected as showing the most robust responses (Figure 3C). This amplitude cutoff was empirically determined through comparison with regression analysis results (Figure 9), representing a balance between response significance and population coverage. The adjustable nature of this cutoff allows for experimental flexibility: lower cutoff prioritize stronger stimulus correlations (Figure 9D) while higher cutoff increase population completeness at the expense of response strength. This flexible approach enables either focused analysis of strongly coupled neurons or comprehensive mapping of sensory-responsive populations, depending on the choice amplitude cutoff.

Following amplitude-based selection, we applied phase cutoffs to ensure temporal alignment

with stimulus presentation. Given the bilateral stimulus design with two distinct temporal intervals (Figure 3A), we expected stimulus-driven neurons to demonstrate phase-locked activity during at least one stimulus block. Phase cutoffs corresponding to stimulus onset and offset were applied, with only neurons falling within these temporal windows are classified as stimulus-driven (Figure 3C). This approach automatically categorized neurons as responding to either left or right phototactic stimulation based on their phase alignment. In the example fish, this combined selection revealed several notable features: (1) a substantial population of neurons showed strong, phase-locked responses to each unilateral stimulation; and (2) the phase distribution of these neurons peaked slightly after the midpoint of stimulus presentation (Figure 3B), suggesting unique temporal synchronization of activity in the phototaxis circuit.

Fourier analysis reveals temporal pattern across neuron activity in different brain regions

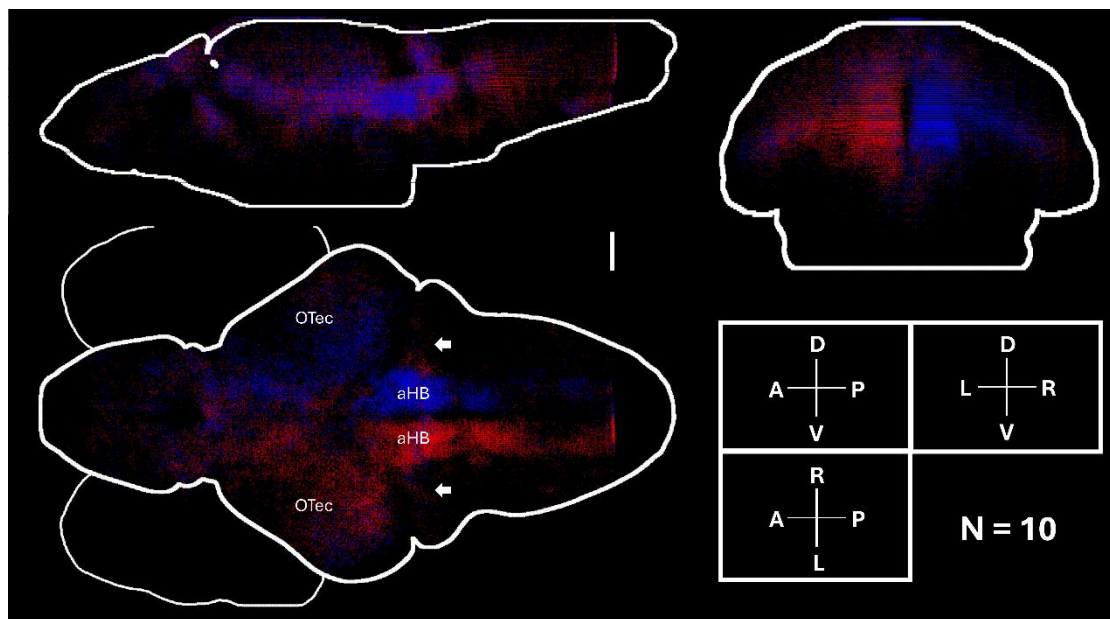


Figure 4. | Spatial Map for Sensory Related Neurons. Spatial distribution of sensory-responsive neurons across multiple zebrafish, color-coded by stimulus preference: red = left-side phototactic stimulus, blue = right-side phototactic stimulus. Scale bar: 50 μ m. Key anatomical regions labeled: OTec (optic tectum), aHB (anterior hindbrain).

To map the anatomical distribution of stimulus-driven neurons, we superimposed Fourier analysis results from 10 zebrafish onto a normalized brain template across coronal, transverse, and sagittal planes (Figure 4). These neurons were distributed throughout the posterior forebrain, midbrain, and hindbrain. Within the midbrain, the Optic Tectum (OTec) responded to contralateral bright stimulus in early phases and contralateral dark stimulus in later phases (Figure 4, OTec; Figure 7, Top left-OTec), consistent with known contralateral eye input⁷ and dark-side responsiveness⁴. The bilateral tectal activity (Figure 7, Top right-OTec) reflects whole-field visual processing, where illumination gradients obtained from input of both eyes guide motor responses⁸.

In hindbrains, the most prominent stimulus-driven neuron clusters are located to the side of

the brain ipsilateral to the side of phototactic stimulation. This demonstrates a synchronized activity pattern between neurons in the hindbrain to stimulus. One previously studied neuron population in this region that responds to phototactic stimulus is the hindbrain oscillator (HBO), which locates in the anterior hindbrain (Figure 4, aHB). This neuron group plays a role in integrating visual input and generating motor response based on illumination information transmitted from anterior parts of the brain. It is known that HBO is activated by ipsilateral bright half-field stimulus⁶. This finding is also aligned with our map of sensory-related neuron based on Fourier analysis, where left side hind brain neurons are associated with left bright half-field stimulus, and vice versa (Figure 4). A few contralaterally activated neuron clusters are also found in the anterior hindbrain, (Figure 4, Arrows), which are the sub-populations of the hind brain oscillator that are induced by visual stimulus from contralateral eye⁶.

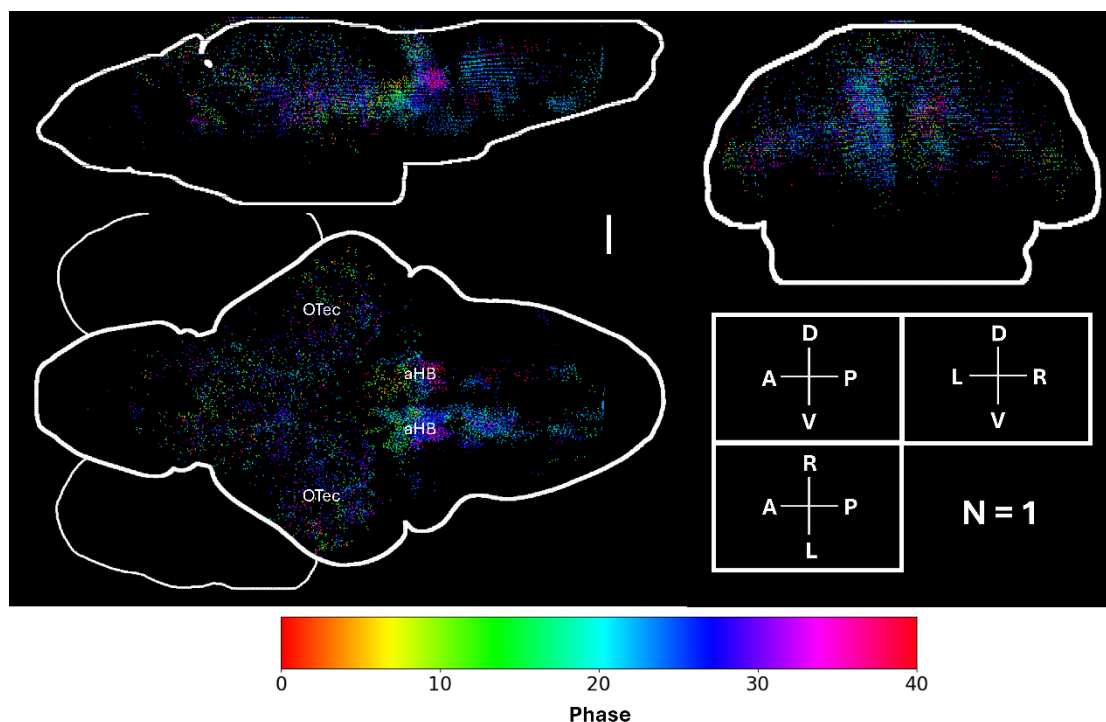


Figure 5. | Phase Map for Sensory Related Neurons. Whole-brain map of stimulus-responsive neurons in a representative zebrafish, with colors indicating normalized phase values relative to stimulus onset (phase 0 = start of stimulus block). Scale bar: 50 μ m. Anatomical abbreviations: OTec (optic tectum), aHB (anterior hindbrain).

Fourier analysis additionally provides phase information encoding the precise timing of neuronal peak responses. To examine activation sequences across populations during phototaxis, we aligned each neuron's phase to its corresponding stimulus onset, creating a normalized phase dimension. While this normalized phase does not represent absolute response latency (as the sinusoidal Fourier approximation may not capture exact neuronal response profiles), it reveals relative activation sequences between neural populations. The normalized phases were then visualized by mapping neurons according to their anatomical positions and coloring them by phase values (Figure 5), enabling spatial-temporal analysis of response patterns across brain regions.

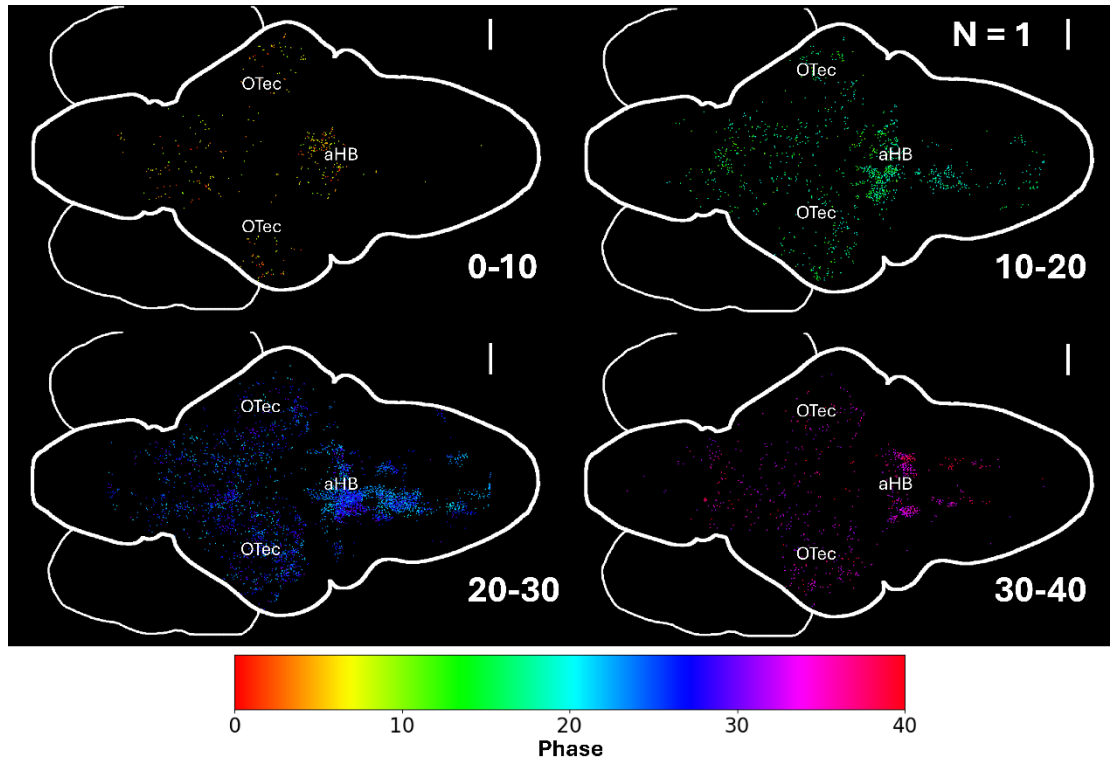


Figure 6. | Ordered Phase Map for Sensory Related Neurons. Whole-brain maps show phase-binned sensory neurons in a sample zebrafish, revealing sequential activation patterns during phototactic stimulation. Four temporal windows are displayed: first quarter (frames 0-10, top left), second quarter (frames 10-20, top right), third quarter (frames 20-30, bottom left), and fourth quarter (frames 30-40, bottom right). Neuron colors represent normalized phase values aligned to stimulus onset (phase 0 = stimulus start). Anatomical labels: OTec (optic tectum), aHB (anterior hindbrain).

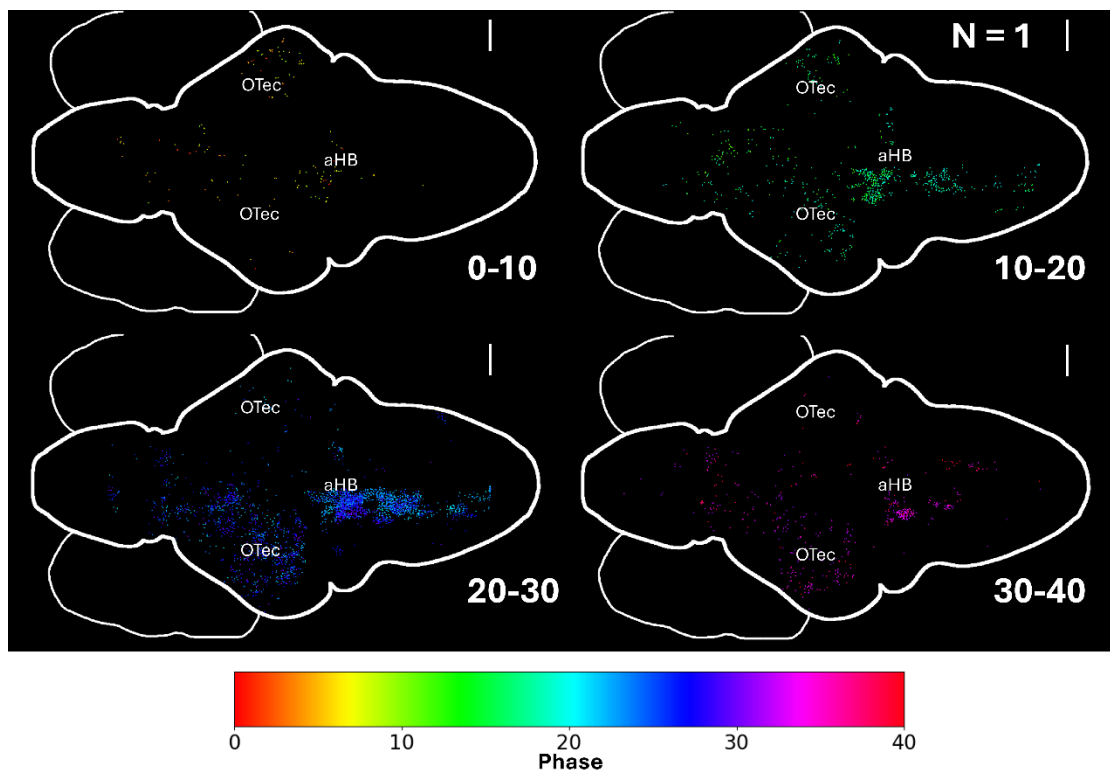


Figure 7. | Ordered Phase Map for left phototactic stimulation associated neurons.

Whole-brain maps show phase-binned sensory neurons associated with left-side phototactic stimulation, revealing sequential activation patterns during stimulation. Four temporal windows are displayed: first quarter response (frames 0-10, top left), second quarter response (frames 10-20, top right), third quarter response (frames 20-30, bottom left), and fourth quarter response (frames 30-40, bottom right). Neuron colors represent normalized phase values aligned to stimulus onset (phase 0 = stimulus start) Anatomical labels: OTec (optic tectum), aHB (anterior hindbrain).

To characterize the temporal sequence of neural activation during phototactic stimulation, we categorized responsive neurons into four sequential phase groups, each representing a quarter of the stimulus cycle (Figure 6). When examining neurons responsive to left-side phototactic stimulation (Figure 7), we observed a distinct spatiotemporal activation pattern across the brain. In the initial phase, the right optic tectum showed strong activation in response to left bright-field stimulation while simultaneous dark stimulation on the right suppressed activity in the left optic tectum (Figure 7, Top left). This immediate contralateral activation pattern is consistent with the known cross-over anatomy of retinal ganglion cell projections⁷.

As the stimulus progressed into the second quarter, we observed the emergence of bilateral optic tectum activation accompanied by increasing activity in the left (ipsilateral) hindbrain (Figure 7, Top right). This bilateral engagement likely reflects the neural circuitry integrating whole-field visual information, which is necessary for detecting the illumination gradients that guide phototactic behavior⁸. The transition into the third quarter revealed a striking shift toward left-side dominance, with robust activation persisting in left midbrain and hindbrain regions while right optic tectum activity diminished significantly (Figure 7, Bottom left).

The final quarter showed reduction of midbrain activity but maintained strong activation in the left hindbrain, particularly within the anterior hindbrain region (Figure 7, Bottom right). This ipsilateral dominance in the hindbrain observed in quarter 2-4 is consistent with the pre-existing results that suggest the HBO is stimulated by ipsilateral illumination⁶. The complete temporal sequence reveals how distributed neural circuits coordinate to transform visual stimuli input from anterior region into corresponding information processing and integration network in the hindbrain during phototactic stimulation.

The observed late-stage ipsilateral dominance in optic tectum presents an interesting phenomenon that extends beyond explanations based solely on retinal wiring or whole-field processing. We propose this pattern may reflect an attention-like mechanism directed toward the dark stimulus, potentially serving as a threat detection response. This interpretation accounts for three key observations: the sustained processing of dark stimulus information in the contralateral (left) optic tectum, the silenced right optic tectum accounting for adaptation of bright stimulus, and the HBO's role in converting processed visual information into avoidance motor response. The precise neural mechanisms underlying this temporal patterning need further investigation, particularly regarding how these phase-specific

activation patterns relate to the higher-order network processing that produces ultimate behavioral output.

Result of Fourier analysis is validated by regression analysis

To validate our Fourier-based identification of stimulus-driven neurons, we performed a direct comparison with regression analysis results from the original dataset⁴. The reference regression analysis used a stimulus regressor constructed by convolving a box-car function representation of the phototactic stimulus with an single exponential kernel modeling GCaMP6f calcium dynamics (half-time = 0.4 s, peak delay = 0.08 s). Neuronal activity showing a correlation coefficient > 0.5 with this regressor were classified as stimulus-driven, establishing our validation baseline.

This comparative approach serves two important purposes: First, it allows quantitative assessment of the Fourier method's accuracy by measuring its agreement with established regression techniques. Second, it provides a framework for evaluating how our phase-amplitude selection criteria (top 10% amplitude cutoff with phase constraints) compared to conventional correlation-based selection. The regression-identified neuron set represents a ground truth, as it directly models both the stimulus timing and the known temporal properties of the calcium indicator.

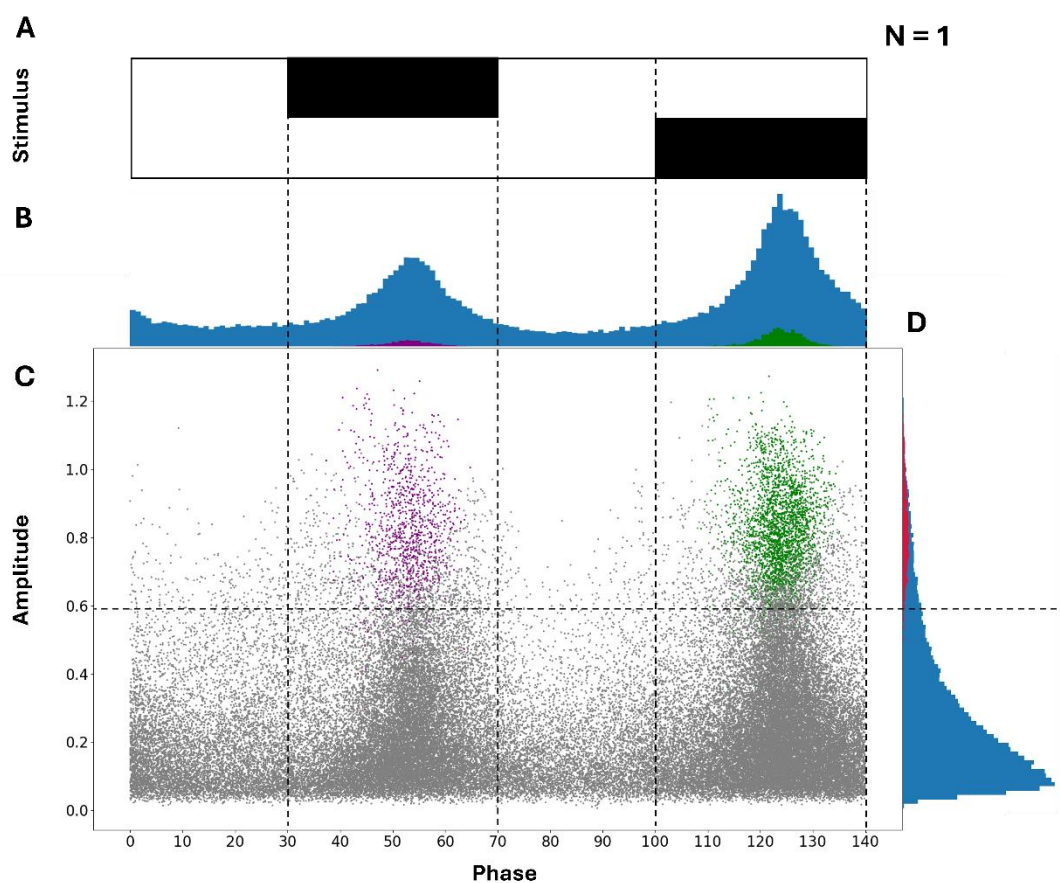


Figure 8. | Distribution of Phase and Amplitude in Neurons in one example fish identified by regression analysis. (A) Single trial showing right (frames 30-70) and left (frames 100-140) phototactic stimuli aligned to neuronal phase values. **(B)** Phase distribution histogram ($n = 66,789$ neurons). **(C)** Amplitude-phase scatter plot showing regression-analysis labeled neurons (purple: right phototactic stimulus correlated [$r > 0.5$]; green: left phototactic stimulus correlated [$r > 0.5$]) with selection cutoffs (horizontal dashed lines: top 10% amplitude; vertical dashed lines: phase cutoffs). **(D)** Amplitude distribution histogram.

To evaluate the correspondence between our Fourier analysis and regression analysis, we examined the phase and amplitude distributions of regression-identified neurons at the stimulus frequency (Figure 8C). The Fourier-transformed regression neurons formed two distinct clusters in the phase-amplitude space, both located in the high-amplitude region of the plot. These clusters were segregated along the phase axis, with each cluster aligning precisely with one of the two stimulus presentation intervals. This distribution pattern strongly validates our Fourier approach, demonstrating that: (1) neurons identified as stimulus-driven through regression analysis indeed show prominent periodic components at the stimulus frequency; and (2) these neurons naturally separate into phase groups corresponding to the different stimulus intervals, just as we observed in our Fourier-based selection. The high-amplitude characteristic of these regression-identified neurons further confirms that our amplitude selection criterion effectively captures neurons with robust stimulus-coupled activity.

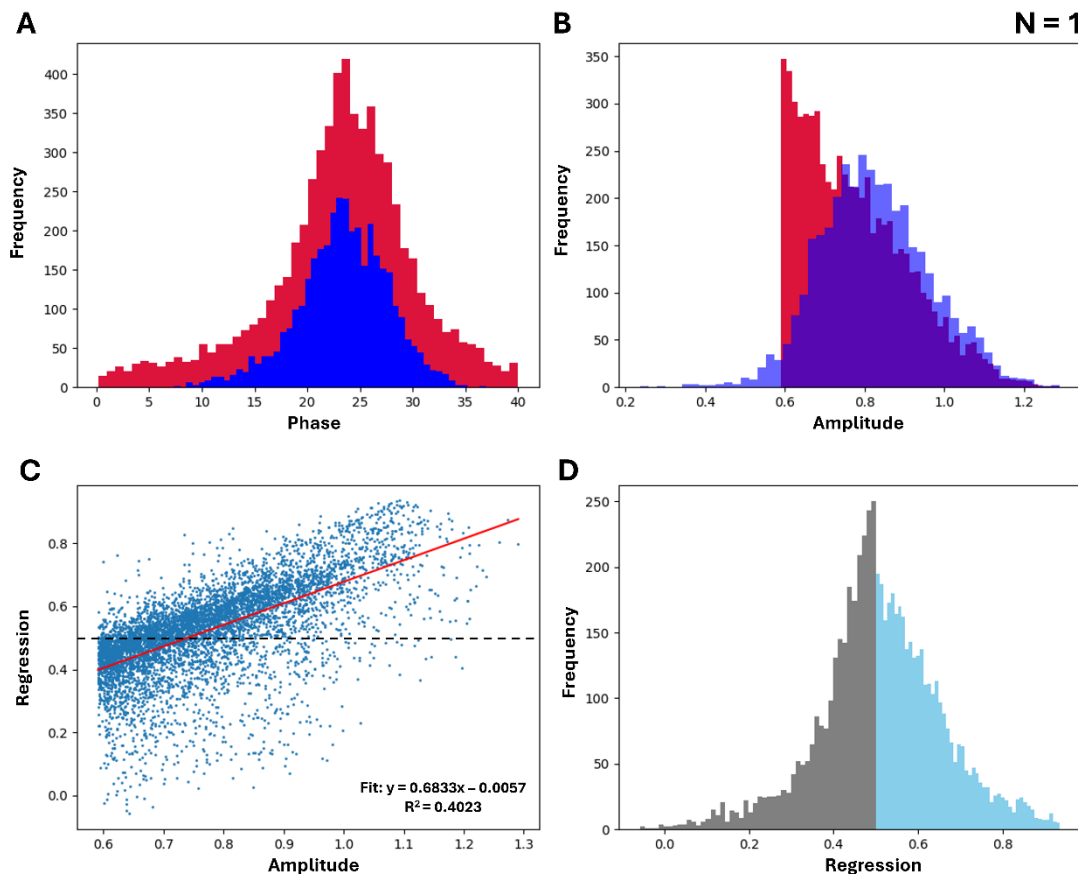


Figure 9. | Comparison between properties of Fourier-analysis and Regression-analysis labeled neurons. (A) Phase distribution comparison between Fourier-identified (red) and regression-identified (blue) stimulus-driven neurons (phases normalized to stimulus block duration). **(B)** Amplitude distributions for Fourier-identified (red) and regression-identified (blue) stimulus-driven neurons. **(C)** Amplitude-correlation relationship in Fourier-identified neurons, showing significant linear fit ($y = 0.6833x - 0.0057$; $R^2 = 0.4023$, $p < 0.001$) with regression threshold ($r > 0.5$, dashed line). **(D)** Regression value distribution of Fourier-identified neurons, colored by significance (blue: $r \geq 0.5$; gray: $r < 0.5$).

Comparison between regression and Fourier analysis also reveals important distinctions in the neuronal populations identified by each method. While both approaches detect neurons with phases aligned to stimulus blocks, the phase distribution of regression-identified neurons is significantly narrower than both the stimulus duration and the Fourier-identified population (Figure 8B; Figure 9A). This difference indicates that regression analysis preferentially selects neurons with peak activity near the midpoint of stimulus blocks, reflecting its limitation in detecting neurons with activity patterns that deviate from the stimulus regressor. Neurons exhibiting transient activation at stimulus onset/offset typically show poor correlation with the regressor and are consequently excluded by regression analysis. However, Fourier analysis successfully identifies these neurons through their strong periodic components at stimulus frequency. This demonstrates Fourier analysis's sensitivity to diverse response patterns that maintain stimulus-locked periodicity.

The concentration of phases near stimulus midpoints in both methods stems from inherent characteristics of their analytical approaches. For regression analysis, this bias emerges because maximally correlated neurons necessarily follow the regressor's temporal profile. In Fourier analysis, this tendency arises because sinusoidal approximations naturally peak at cycle midpoints - a property confirmed by applying DFT to the stimulus regressor itself, which yields a phase precisely at the center of each stimulus block. These findings highlight how each method reveals distinct aspects of stimulus processing: while regression effectively captures neurons tracking stimulus dynamics, Fourier analysis provides a more comprehensive view of periodic responses, including transient and offset responses that may represent important function in phototaxis circuits.

We next compared the amplitude distributions of neurons identified through regression analysis versus Fourier analysis (Figure 9B). The amplitude distribution of Fourier-identified neurons exhibited a distinct horizontal asymptote, a direct consequence of our amplitude cutoff in selection criteria. In contrast, regression-identified neurons showed a more symmetrical amplitude distribution. The comparison revealed several important patterns. In the high-amplitude range, we observed major overlaps between neurons identified by both methods (Figure 9B, purple region), indicating strong agreement for neurons with robust stimulus-coupled responses. However, at lower amplitudes, Fourier analysis captured additional neurons - particularly those phase-shifted neurons with peak activity near stimulus block edges - that were missed by regression analysis due to their temporal response profiles. Conversely, regression analysis identified some low-amplitude neurons (Figure 9B, blue left

region) that fell below our Fourier amplitude cutoff yet showed high correlation with the stimulus regressor. This discrepancy points out a limitation of correlation-based approaches: while sensitive to temporal pattern matching, they are insensitive to response magnitude. Two neurons with identical response patterns but different response strengths would yield identical correlation coefficients despite their differing activation levels. This observation highlights the advantage of Fourier-derived amplitude measurements, which provide direct quantitative information about response strength that is lost in pure correlation analyses. The amplitude parameter thus offers a more physiologically meaningful metric of neuronal engagement, as it reflects the actual magnitude of activity rather than just temporal synchronization with the stimulus.

To investigate the relationship between activity amplitude and stimulus correlation, we computed the correlation coefficients of Fourier-identified neurons with the stimulus regressor and plotted them against their corresponding amplitudes (Figure 9C). The analysis revealed a positive relationship ($r^2 = 0.4$, $p < 0.001$ by linear regression), demonstrating that neurons with higher amplitude responses tend to show stronger correlation with the stimulus pattern. This finding further supports the utility of amplitude as a metric for identifying stimulus-responsive neurons. The correlation coefficient distribution among Fourier-identified neurons (Figure 9D) showed that approximately half fell below the regression threshold ($r > 0.5$). These subthreshold neurons correspond to the phase-shifted population described earlier - neurons that respond to the stimulus but with temporal profiles not perfectly synchronized to the stimulus timing. Importantly, while these neurons are overlooked by regression analysis, Fourier analysis successfully captures them through their strong periodic components at stimulus frequency.

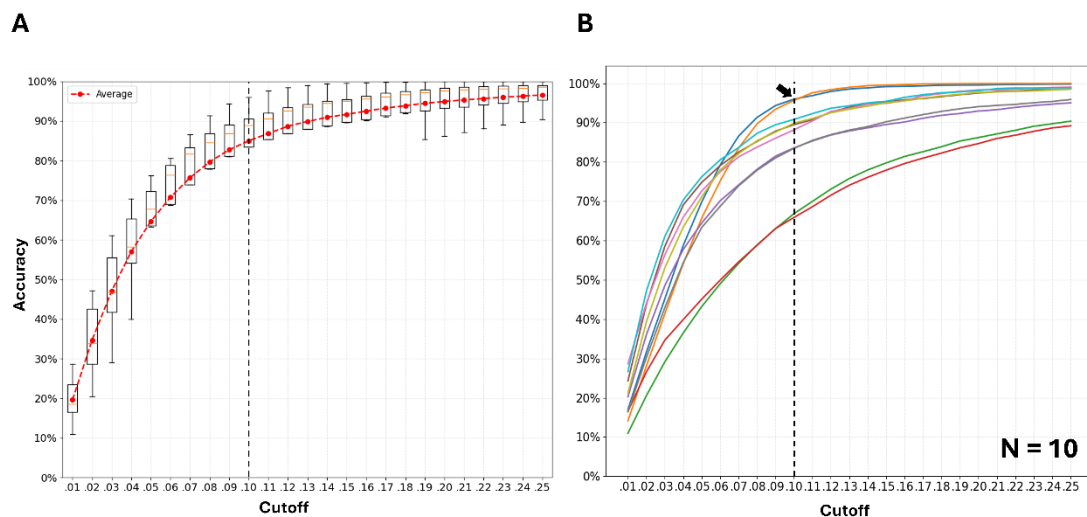


Figure 10. | Accuracy of Fourier analysis at different cutoff levels. (A) Accuracy of Fourier analysis (defined as overlap with regression-identified neurons) across amplitude cutoff levels (boxplot). Dashed line indicates selected cutoff used in the paper (top 10%). **(B)** Accuracy trajectories for individual zebrafish (n=10), with dashed line marking chosen cutoff and arrow indicating example fish performance.

In this paper, we employed a consistent amplitude cutoff of 0.1, selecting the top 10% of neurons with the highest response amplitudes as stimulus driven. This cutoff represents a balance between statistical significance and population completeness. Increasing the cutoff includes more lower-amplitude neurons at the cost of reduced response specificity; while decreasing it focuses only on the strongest responders but risks incomplete sampling of relevant neuronal populations.

To systematically determine an appropriate cutoff, we evaluated cutoff values ranging from 0.01 to 0.25 in 0.01 increments across 10 experimental subjects (Figure 10), quantifying accuracy as the percentage overlaps with regression-identified neurons ($\text{Accuracy} = \text{number of neurons identified by both method} / \text{number of neurons identified by regression analysis alone}$). The 0.1 cutoff yielded an average accuracy of 83% across all subjects (Figure 10A), reaching 95% in the example fish shown in the paper (Figure 10B, arrow). Importantly, analytical accuracy showed strong dependence on the number of stimulus trials, with performance declining for datasets containing fewer stimulus trials. These results demonstrate that while our selected cutoff provides robust performance for typical experimental conditions, researchers should carefully adjust this parameter based on their specific experiment paradigm and analytical goals - whether prioritizing response reliability through lower cutoff or population completeness through higher cutoff.

Discussion

We present a Fourier transformation approach for analyzing neuronal activity during periodic perturbations, such as repeated sensory stimulation or stereotyped motor output. This method capitalizes on the principle that stimulus- or motor-related neurons exhibit activity patterns synchronized to the perturbation frequency. Discrete Fourier Analysis (DFT) decomposes neuronal signals into their frequency components, isolating periodic responses through sinusoidal approximation (Figure 1B). The component at the stimulus frequency characterizes a neuron's periodic response, described by three key parameters: (1) frequency (locked to the stimulus), (2) amplitude (quantifying response strength in $\Delta F/F$), and (3) phase (encoding response timing). These parameters enable systematic classification of stimulus-correlated neurons when evaluated against stimulus-derived selection criteria.

We implemented this Fourier-based approach on a publicly available whole-brain light-sheet imaging dataset of zebrafish larvae undergoing periodic phototactic stimulation⁴. The stimulus periodicity enabled decomposition of neuronal activity into frequency components via DFT. From each neuron's Fourier spectrum, we: (1) extracted the stimulus-frequency component, (2) selected the top 10% amplitude responses, and (3) classified these neurons by response timing (phase) into left- or right-stimulus groups. This pipeline successfully identified spatially distributed, stimulus-locked neural populations (Figure 3).

By aggregating Fourier-identified neurons across 10 fish, we generated whole-brain activation maps revealing spatially distributed populations associated with left/right phototactic stimulation. These maps confirmed robust stimulus-locked activity in known

visual processing regions, including the optic tectum in midbrain and hindbrain oscillator in anterior hindbrain⁶. The spatial organization and laterality patterns observed through our Fourier-based approach align well with established anatomical and functional studies of phototaxis circuits.

Fourier analysis revealed precise temporal organization in phototaxis-related activity across brain regions (Figures 5-6). The response sequence begins with contralateral optic tectum activation, followed by bilateral midbrain engagement - consistent with whole-field illumination processing⁸. During later phases, activity shifts ipsilaterally, potentially reflecting higher-order modulation like attention orientated toward the dark stimulus. Concurrently, activity in hindbrain oscillators emerge during bilateral tectal activation and persist throughout stimulation, confirming its role as a visual input integrator⁶. This spatiotemporal progression - from initial contralateral visual processing in tectum⁷ to sustained hindbrain integration⁶ - reveals how distributed circuits coordinate to integrate visual input through multiple brain regions. The observed laterality shifts may represent adaptive mechanisms in phototaxis circuits.

We validated our Fourier approach by comparing it with established regression methods from the original dataset⁴. Regression-identified neurons are densely clustered in amplitude-phase space after DFT (Figure 8), demonstrating the potential for precise identification by suitable selection strategy. When applying our stimulus-derived selection criteria (Figure 1A), Fourier analysis captured 83% of regression-identified neurons on average across 10 fish (Figure 10B), confirming its effectiveness for detecting stimulus-related activity.

Our comparison of regression and Fourier analysis revealed important differences in their sensitivity to response properties (Figure 9A,B). The phase distribution of regression-identified neurons was notably narrower and centered around stimulus midpoints compared to Fourier analysis (Figure 9A), reflecting regression's inability to detect neurons with temporally shifted responses. These phase-shifted neurons, while strongly periodic, show reduced correlation with the stimulus regressor due to their temporal offset. In the amplitude domain, both methods showed strong agreement for high-amplitude responses, representing neurons with robust, stimulus-locked activity. However, Fourier analysis uniquely identified high-amplitude neurons with phase-shifted responses that missed regression. Conversely, regression captured some low-amplitude neurons that fell below our Fourier cutoff (Figure 9B). This discrepancy highlights the insensitivity to activity amplitude in linear regression, whereas Fourier analysis preserves this important amplitude information.

Analysis of stimulus correlation in Fourier-identified neurons revealed a significant positive relationship between amplitude and correlation to stimulus (Figure 9C), though approximately half fell below the conventional regression threshold ($r > 0.5$; Figure 9D). This difference arises because phase constraints alone cannot perfectly predict correlation strength - even with appropriate phase cutoffs, no single amplitude cutoff exclusively captures high-correlation neurons. The subthreshold population primarily consists of phase-shifted neurons that, despite their lower regression values, exhibit two key signatures of stimulus coupling: (1)

strong amplitude responses and (2) precise frequency locking to the stimulus. These characteristics suggest that these neurons are likely to participate in stimulus processing. Although their activity is dissimilar to the stimulus pattern, these phase-shifted neurons may still play important functional roles in phototaxis behavior.

While phase cutoffs were determined by stimulus timing, amplitude cutoffs required empirical optimization. We selected the top 10% amplitude responses (cutoff = 0.1) as it achieved 83% accuracy relative to regression analysis across 10 fish (95% in the example fish; Figure 10B). Systematic evaluation of cutoffs (0.01-0.25 range) revealed an inherent tradeoff: higher cutoff improved completeness but included weaker responses, while lower cutoff prioritized strong responders at the cost of reduced sample size and anatomical coverage. Notably, analytical accuracy scaled with trial number, emphasizing that optimal cutoff selection should consider both experimental parameters (trial count) and research objectives (significance vs. completeness), rather than adopting a universal value.

The strength of Fourier analysis is that it extracts three biologically meaningful parameters from neural activity: frequency (stimulus locking), amplitude (response strength), and phase (response timing). Unlike regression methods, it specifically quantifies response magnitude through amplitude while also capturing temporal relationships via phase. This provides a new perspective in studying brain dynamics and enables investigation of information flow through neural circuits by revealing activation sequences across brain regions. Although our current 2Hz temporal resolution limits observations to broad temporal patterns, higher sampling rates could potentially resolve millisecond-scale dynamics in neural circuits. Another key strength is the method's ability to detect temporally shifted responses that regression analysis misses, including neurons with transient onset/offset activity patterns. By identifying these phase-specific responses, Fourier analysis provides a more comprehensive view of stimulus-related activity beyond what correlation-based methods can reveal.

Several important limitations should be considered when applying Fourier analysis to calcium imaging data. First, the method relies on two layers of approximation: (1) calcium fluorescence ($\Delta F/F$) as an indirect neural activity measure⁴, and (2) approximation of neural activity through sinusoidal functions that may not perfectly capture true neural response dynamics. We expect each layer of approximation reduces the validity of the result; thus, it raises questions about how precisely the extracted parameters in the end reflect actual neuronal activity. Regard this limitation, we recommend validating the result of Fourier analysis with complementary time-domain analyses like cross-correlation. Additionally, our current implementation focuses solely on the stimulus-frequency component. Incorporating additional frequency components could improve signal reconstruction accuracy and potentially serve as a dimension reduction technique for neural time series - a possible direction for future methodological development.

Another fundamental constraint of Fourier analysis is its requirement for periodic neural perturbations, limiting application to specific stimulus paradigms. This approach cannot accommodate unpredictable or aperiodic motor outputs, as commonly observed in

behavioral experiments. However, this generalizability limitation can be addressed through careful experimental design incorporating repeated periodic stimuli. And the method remains well-suited for studying organized, rhythmic behaviors like optokinetic responses⁹. When applied carefully to appropriate paradigms, Fourier analysis serves as a powerful complementary tool in computational neuroscience, offering unique insights when used alongside other analytical methods.

Methodology

Discrete Fourier Transformation of neural activity data

Neural activity time series were analyzed using the Discrete Fourier Transform (DFT) implemented via SciPy's Fast Fourier Transform (FFT) module. For a given neural trace x_t of length t frames, the transformed output Y was computed as:

$$Y = \left\{ Y_k \mid Y_k = \sum_{n=1}^t x_n \cdot e^{-2\pi i \frac{kn}{t}}, \text{ for } 1 \leq k \leq t \right\}$$

For a neural activity time series x_t with t frames, the DFT yields complex-valued components Y_k ($1 \leq k \leq t$). Analysis was restricted to the real-frequency domain components, corresponding to:

$$m = \begin{cases} \frac{t}{2}, & \text{for even } t \\ \frac{t-1}{2}, & \text{for odd } t \end{cases}$$

where only Y_k for $k \in [1, m]$ were retained for subsequent analysis. This preserves all unique frequency components while eliminating redundant negative frequencies in the real-valued transform.

The frequency basis F for the Fourier-transformed components was computed as:

$$F = \left\{ F_k \mid F_k = \frac{k}{t}, \text{ for } 1 \leq k \leq m \right\}$$

The wavelength λ (in frames) for each frequency component is given by the reciprocal relationship:

$$\lambda = \frac{1}{F} = \left\{ \lambda_k \mid \lambda_k = \frac{t}{k}, \text{ for } 1 \leq k \leq m \right\}$$

The amplitude spectrum A was computed from the complex Fourier coefficients as:

$$A = |Y| = \left\{ A_k \mid A_k = |Y_k| \cdot \frac{2}{t}, \text{ for } 1 \leq k \leq m \right\}$$

The phase spectrum θ was computed from the complex Fourier coefficients as:

$$\theta = \arg(Y) = \left\{ \theta_k \mid \theta_k = \arg(Y_k) \text{ for } 1 \leq k \leq m \right\}$$

The phase angles were converted to stimulus-relative time frames using:

$$P = \left\{ P_k | P_k = \lambda_k \cdot \frac{-\theta_k \bmod 2\pi}{2\pi}, \text{ for } 1 \leq k \leq m \right\}$$

Sinusoidal approximation of neural activity at given frequency.

The sinusoidal approximation of neural activity at a specific frequency $F_k \in F$ (where $F_k = k/t$ represents the k -th frequency component) is reconstructed using its corresponding amplitude A_k and phase P_k . The time-domain approximation $S(F_k, A_k, P_k)$ is given by:

$$S(F_k, A_k, P_k) = A_k \cdot \cos(2\pi F_k \cdot (x - P_k)) \text{ for } x \text{ in } [0, t]$$

This formulation ensures precise alignment with the stimulus periodicity while preserving the original signal's amplitude scaling (A_k in $\Delta F/F$ units). The approximation captures the dominant oscillatory component at F_k , facilitating analysis of stimulus-locked neural dynamics. For multi-frequency reconstruction, the full signal can be approximated by summing over all frequencies of interest.

Identification of Stimulus-Driven Neuron by Fourier analysis

To identify stimulus-driven neurons, we first characterized the periodicity of the phototactic stimulus by computing the Discrete Fourier Transform (DFT) of a box-car function representing the stimulus train, yielding the fundamental stimulus frequency (F_{stim}) and temporal stimulation window (T_{stim}).

For each neuron's calcium imaging trace, we performed DFT and extracted the component at F_{stim} by selecting the frequency bin k where $F_k = F_{stim}$, obtaining the amplitude (A_k) and phase (P_k) of the neuron's response at the stimulus frequency.

Neurons were then filtered through two selection criteria: (1) amplitude thresholding, where only neurons with amplitude above the cutoff (the top 10% of response amplitudes) were retained, and (2) phase alignment, where the response phase P_k had to fall within the active stimulation window T_{stim} . This dual-threshold approach ensured the selection of neurons that were both strongly modulated by the stimulus and precisely time-locked to the stimulus period, while excluding weakly responding or poorly aligned cells. The phase alignment further allowed classification of neurons based on their specific response timing within the stimulus cycle.

Identification of Stimulus-Driven Neuron by Regression analysis:

To enable comparison with conventional methods, we replicate a regression analysis pipeline⁴. First, we constructed a stimulus regressor by convolving a box-car function representing the phototactic stimulus timing with a single-exponential kernel modeling GCaMP6f calcium

dynamic ($\tau = 0.4$ s half-time, 0.08 s peak delay). For each neuron's fluorescence trace, we then computed the Pearson correlation coefficient (r) between the observed $\Delta F/F$ signal and the convolved regressor. Neurons demonstrating significant stimulus coupling ($r > 0.5$ threshold) were classified as stimulus driven.

Validation Metric for Fourier Analysis Performance

To quantify the agreement between Fourier and regression methods, we calculated the percentage accuracy of Fourier-identified neurons relative to the regression ground truth. For a given set of neurons identified by Fourier analysis (set A) and regression analysis (set B), the accuracy metric is defined as:

$$\text{Percentage Accuracy} = \frac{|A \cap B|}{|B|} \times 100\%$$

Where:

- $|A \cap B|$ = number of neurons identified by both methods
- $|B|$ = total neurons identified by regression analysis ($r > 0.5$)

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